

Removing decimal power/fraction power both sides

$$\left\{ 2 - \left(\frac{dy}{dx} \right)^2 \right\}^{\frac{3}{5}} = \frac{d^2y}{dx^2}$$

Raise power of 5 both sides,

$$\left\{ 2 - \left(\frac{dy}{dx} \right)^2 \right\}^{\frac{3}{5} \times 5} = \left(\frac{d^2y}{dx^2} \right)^5$$

$$\Rightarrow \left\{ 2 - \left(\frac{dy}{dx} \right)^2 \right\}^3 = \left(\frac{d^2y}{dx^2} \right)^5$$

Now differential equation is free from decimal or fraction power.

So, Order = 2

Degree = 5

76. Option (a) is correct.

$$\frac{dy}{dx} = 2e^x y^3,$$

$$\Rightarrow \frac{dy}{y^3} = 2e^x dx$$

Integrating both sides w.r.t. x

$$\frac{-1}{2y^2} = 2e^x + C \quad \dots(i)$$

$$y(0) = \frac{1}{2}, \quad \text{Substitute in (i)}$$

$$-2 = 2 + C$$

$$\Rightarrow C = -4$$

The solution is:

$$\frac{-1}{2y^2} = 2e^x - 4$$

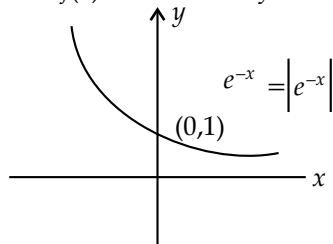
$$\Rightarrow 4y^2(e^x) - 8y^2 = -1$$

$$\Rightarrow 4y^2(2 - e^x) = 1$$

77. Option (d) is correct.

$$\text{Given, } p = \int_a^b f(x) dx \text{ and } q = \int_a^b |f(x)| dx$$

Since $f(x) = e^{-x}$ is always +ve $\forall x \in \mathbb{R}$



$$|f(x)| = |e^{-x}| = e^{-x} = f(x)$$

$$\therefore \int_a^b f(x) dx = \int_a^b |f(x)| dx$$

$$\therefore p = q$$

78. Option (a) is correct.

$$\text{Let } I = \int_0^{\frac{\pi}{2}} \frac{a + \sin x}{2a + \sin x + \cos x} dx \quad \dots(i)$$

$$I = \int_0^{\frac{\pi}{2}} \frac{a + \cos x}{2a + \cos x + \sin x} dx \quad \dots(ii)$$

$$\left[\int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$$

Adding equation (i) and (ii)

$$2I = \int_0^{\frac{\pi}{2}} \frac{2a + \sin x + \cos x}{2a + \sin x + \cos x} dx$$

$$= \int_0^{\frac{\pi}{2}} dx = \frac{\pi}{2}$$

$$\Rightarrow I = \frac{\pi}{4}$$

79. Option (d) is correct.

$$\text{Let } f(x) = \frac{16x^3}{3} - 4bx^2 + x$$

$$f'(x) = 16x^2 - 8bx + 1$$

Since $f(x)$ has neither maximum nor minimum, then $f'(x) \neq 0$

$$f'(x) > 0 \text{ or } f'(x) < 0$$

(but here $a > 0$ for $f'(x)$ so not possible)

for $f'(x) > 0$

Discriminant of $f'(x)$ is $D < 0$ and $a > 0$

$$D = 64b^2 - 64 < 0$$

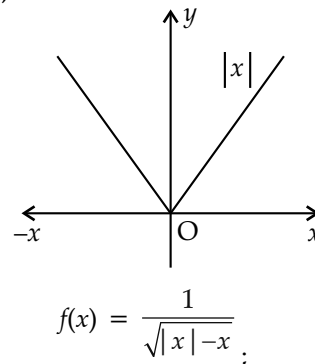
$b^2 - 1 < 0$ and b is non-negative as $b \geq 0$

$b \in (-1, 1)$ and $b \geq 0$

$\therefore b \in [0, 1)$

$\therefore 0 \leq b < 1$

80. Option (a) is correct.



Domain of $f(x)$, $|x| - x > 0$

i.e., $|x| < x$

$\Rightarrow x \in (-\infty, 0)$